

CSC 211: Computer Programming

Number Systems, Further look into DataTypes

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Original design and development by Dr. Marco Alvarez

Administrative Notes

Administrative notes

- A00
- MC01

Number Systems

Number systems

- A way to represent numbers
 - ✓ numbers are expressed in a certain **base**
- Why study number systems in CS?
 - ✓ to understand data representation
- Examples of number systems
 - ✓ binary
 - ✓ decimal
 - ✓ octal
 - ✓ hexadecimal

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Positional number systems

assuming base **b**:

$$\dots d_2 b^2 + d_1 b^1 + d_0 b^0 + d_{-1} b^{-1} + d_{-2} b^{-2} \dots$$

$$43.23 = 4 \cdot 10^1 + 3 \cdot 10^0 + 2 \cdot 10^{-1} + 3 \cdot 10^{-2}$$

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Decimal number system

- Base 10
- Symbols

0 1 2 3 4 5 6 7 8 9

$$456 = 4 \cdot 10^2 + 5 \cdot 10^1 + 6 \cdot 10^0$$

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Binary number system

- Base 2
- Symbols

0 1

Most
Significant Bit

Least
Significant Bit

$$1010 = (1 \cdot 2^3) + (0 \cdot 2^2) + (1 \cdot 2^1) + (0 \cdot 2^0)$$



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Binary to Decimal?

1 0 0 1 0 1 0 0 0

2^0	2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8
1	2	4	8	16	32	64	128	256

C

Try these ..

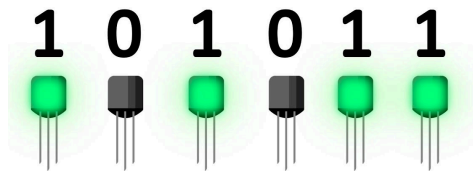
$$\begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

What is a **bit**? What is a **byte**?

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Bits and computers

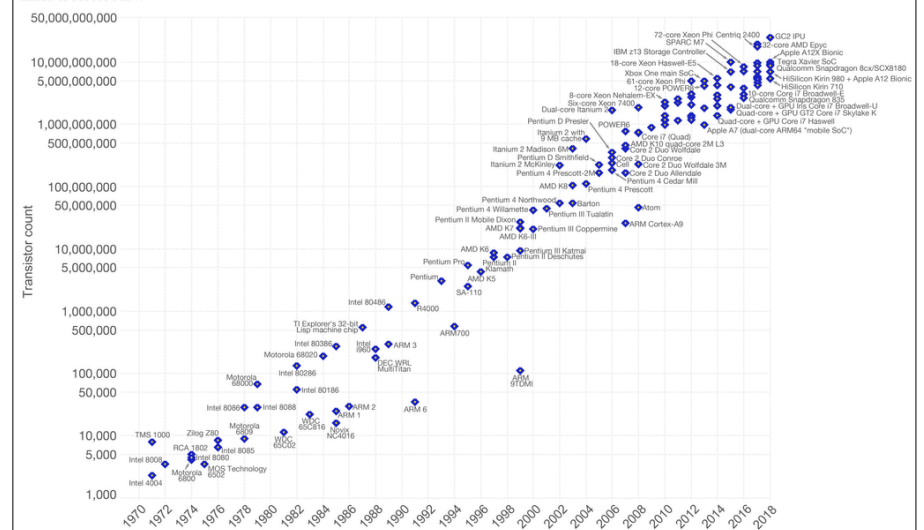
- A bit can only have two values (states)
 - easy to embed into physical devices
- **Transistor**
 - processors have billions of transistors
 - transistors can be switched **on** and **off**



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Moore's Law – The number of transistors on integrated circuit chips (1971-2018)

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important as other aspects of technological progress – such as processing speed or the price of electronic products – are linked to Moore's law.



Data source: Wikipedia (https://en.wikipedia.org/wiki/Transistor_count)
The data visualization is available at [OurWorldinData.org](https://ourworldindata.org). There you find more visualizations and research on this topic

Licensed under [CC-BY-SA](#) by the author Max Roser.

Decimal to other bases

- Repeatedly divide by **base**

- ✓ collect remainders
- ✓ output in reverse order

57_{10}

$\checkmark 57 / 2 = 28 \text{ R } 1$
 $\checkmark 28 / 2 = 14 \text{ R } 0$
 $\checkmark 14 / 2 = 7 \text{ R } 0$
 $\checkmark 7 / 2 = 3 \text{ R } 1$
 $\checkmark 3 / 2 = 1 \text{ R } 1$
 $\checkmark 1 / 2 = 0 \text{ R } 1$

111001_2

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Decimal to other bases

$$57 - 32 = 25$$

$$25 - 16 = 9$$

$$9 - 8 = 1$$

$$1 - 1 = 0$$

57_{10}

111001_2

$$\frac{1}{2^5} \quad \frac{1}{2^4} \quad \frac{1}{2^3} \quad \frac{0}{2^2} \quad \frac{0}{2^1} \quad \frac{1}{2^0}$$

2^0	2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8
1	2	4	8	16	32	64	128	256

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Hexadecimal number system

- Base 16

- Symbols

0 1 2 3 4 5 6 7 8 9 A B C D E F

$$4A1C = (4 \cdot 16^3) + (10 \cdot 16^2) + (1 \cdot 16^1) + (12 \cdot 16^0)$$

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Hexadecimal to decimal

1 D Bx16

A 0 1 0 F

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Binary to hexadecimal

Hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Bin	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
Dec	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Oct	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17

10011101
11010011
11111111

Humans think in **base 10**. Computers think in **base 2**.
Humans use **base 16** to easily manipulate data in **base 2**.

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Color codes

Shades of yellow color chart

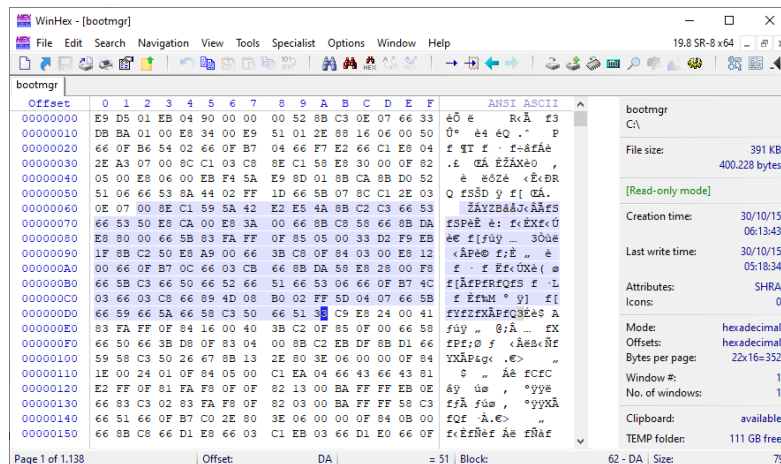
Color	HTML / CSS Color Name	Hex Code #RRGGBB	Decimal Code (R,G,B)
	lightyellow	#FFFFE0	rgb(255,255,224)
	lemonchiffon	#FFFACD	rgb(255,250,205)
	lightgoldenrodyellow	#FAFAD2	rgb(250,250,210)
	papayawhip	#FFEFD5	rgb(255,239,213)
	moccasin	#FFE4B5	rgb(255,228,181)
	peachpuff	#FFDAB9	rgb(255,218,185)
	palegoldenrod	#EEE8AA	rgb(238,232,170)
	khaki	#F0E68C	rgb(240,230,140)
	darkkhaki	#BDB76B	rgb(189,183,107)
	yellow	#FFFF00	rgb(255,255,0)
	olive	#808000	rgb(128,128,0)
	greenyellow	#ADFF2F	rgb(173,255,47)
	yellowgreen	#9ACD32	rgb(154,205,50)

What is the color code of 'greenyellow' in binary?

https://www.rapidtables.com/web/color/Yellow_Color.html

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Forensic Analysis



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31 oct = 25 dec?

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Going back to C++ ...

Integer literals in C++

```
int d = 42;
int o = 052;
int x = 0x2a;
int X = 0X2A;
int b = 0b101010; // C++14
```

- ✓ **decimal-literal** is a non-zero decimal digit (1, 2, 3, 4, 5, 6, 7, 8, 9), followed by zero or more decimal digits (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)
- ✓ **octal-literal** is the digit zero (0) followed by zero or more octal digits (0, 1, 2, 3, 4, 5, 6, 7)
- ✓ **hex-literal** is the character sequence `0x` or the character sequence `0X` followed by one or more hexadecimal digits (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, a, A, b, B, c, C, d, D, e, E, f, F)
- ✓ **binary-literal** is the character sequence `0b` or the character sequence `0B` followed by one or more binary digits (0, 1)

https://en.cppreference.com/w/cpp/language/integer_literal

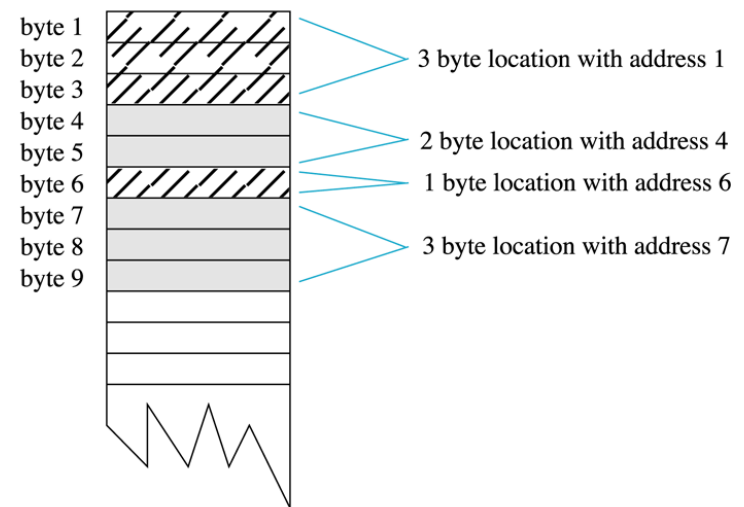
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Type	Size in bits	Format	Value range	
			Approximate	Exact
character	8	signed		-128 to 127
		unsigned		0 to 255
	16	unsigned		0 to 65535
integer	32	unsigned		0 to 1114111 (0x10ffff)
	16	signed	$\pm 3.27 \cdot 10^4$	-32768 to 32767
		unsigned	$0 \text{ to } 6.55 \cdot 10^4$	0 to 65535
	32	signed	$\pm 2.14 \cdot 10^9$	-2,147,483,648 to 2,147,483,647
		unsigned	$0 \text{ to } 4.29 \cdot 10^9$	0 to 4,294,967,295
	64	signed	$\pm 9.22 \cdot 10^{18}$	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
floating point		unsigned	$0 \text{ to } 1.84 \cdot 10^{19}$	0 to 18,446,744,073,709,551,615
	32	IEEE-754	- min subnormal: $\pm 1.401,298,4 \cdot 10^{-45}$ - min normal: $\pm 1.175,494,3 \cdot 10^{-38}$ - max: $\pm 3.402,823,4 \cdot 10^{38}$	- min subnormal: $\pm 0x1p-149$ - min normal: $\pm 0x1p-126$ - max: $\pm 0x1.fffffep+127$
	64	IEEE-754	- min subnormal: $\pm 4.940,656,458,412 \cdot 10^{-324}$ - min normal: $\pm 2.225,073,858,507,201,4 \cdot 10^{-308}$ - max: $\pm 1.797,693,134,862,315,7 \cdot 10^{308}$	- min subnormal: $\pm 0x1p-1074$ - min normal: $\pm 0x1p-1022$ - max: $\pm 0x1.fffffffffffffp+1023$

<https://en.cppreference.com/w/cpp/language/types>

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Memory Locations and Bytes



from: Problem Solving with C++, 10th Edition, Walter Savitch

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